25900553 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACPES201

LIFESTYLE DISEASES & PHYSICAL ACTIVITY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Temperature unit used in science [K] / [CO4]
a). °F b). °C c). K d). J

All three are correct

2. Health problems that can increase hypertension risk include: [K] / [CO3]

a) Diabetes and high cholesterol

b) Asthma and arthritis

c) Tuberculosis and malaria

d) Migraine and epilepsy

3. Which vitamin is not fat-soluble? [K] / [CO6]
a). Vitamin A **b). Vitamin C** c). Vitamin K d). Vitamin D

4. Which hormone is usually high in women with PCOS? [K] / [CO3]
a). Estrogen b). Progesterone **c). Androgens** d).Oxytocin

5. Post-workout, athletes should drink how much per pound lost? [U] / [CO5]

   a). 5–10 ounces b). 11–15 ounces

   c). 16–24 ounces d). 30–40 ounces

[1x5 = 5]

Part B

SHORT ANSWER QUESTIONS

Answer any 5 questions

Each question carries 2 marks

6. How does physical fitness affect mental well-being ? [U] / [CO1]

   Physical fitness reduces stress, improves mood, enhances self-esteem, and promotes better sleep — all of which support mental well-being.

7. Name two psychological consequences of an unhealthy lifestyle. [An] / [CO2]

   Psychological consequences: Stress, anxiety, depression, or low self-esteem

8. Define diabetes mellitus [U] / [CO2]
   Diabetes mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels due to insufficient insulin production or utilization

9. Why is it important to sit and lift objects properly? [U] / [CO3]
   Proper sitting and lifting prevent strain on the spine and reduce the risk of back injury or musculoskeletal disorders.

10. Give two preventive measures for Goiter (any two) [E] / [CO6]
    - Use iodized salt daily.
    - Eat iodine-rich foods like fish, eggs, and milk.
    - Avoid goitrogenic foods such as cabbage and cauliflower.
    - Maintain a balanced diet with minerals like selenium and zinc.
    - Go for regular thyroid check-ups.
    - Promote awareness about iodine deficiency and its prevention.

11. List two purposes of lunges. (Any two) [K] / [CO5]

    - Strengthen leg muscles – especially quadriceps, hamstrings, and glutes.
    - Improve balance and coordination through single-leg stability.
    - Enhance core strength by engaging abdominal and lower back muscles.
    - Increase flexibility in hips and lower body joints.

- Promote better posture by strengthening stabilizing muscles.
- Support functional fitness useful for daily movements like walking, climbing, and bending.

12. What do you mean by lifestyle diseases [U] / [CO1]

Lifestyle diseases are health problems that develop due to unhealthy habits and daily behaviors such as poor diet, physical inactivity, stress, smoking, and alcohol use.
Examples include diabetes, obesity, hypertension, and heart disease.

[2x5 = 10]

Part C

SHORT ESSAY QUESTION

Answer any 4 questions

Each question carries 5 marks

13. Explain how to calculate Target Heart Rate (THR) with an example. [A] / [CO6]

Target Heart Rate (THR) is the optimal range of heartbeats per minute (bpm) during exercise that ensures the body is working hard enough to gain cardiovascular benefits, but not so hard as to cause strain or overexertion.
Formula for Target Heart Rate
There are two commonly used methods:
Basic Formula:
THR = (Maximum Heart Rate × Desired Intensity %)
Karvonen Formula (more accurate):
THR = [(220 − Age) − Resting Heart Rate] × Intensity + Resting Heart Rate

Example Calculation : For a 20-year-old individual, (Students may write any example and use any formula. The one given below is just an example using the basic formula.)
Basic formula : THR = Maximum Heart Rate × Desired Intensity%
Maximum Heart Rate = 220 − Age
Maximum HR = 220 − 20 = 200 bpm.

- *50%: 200 × 0.50 = 100 bpm*
- *60%: 200 × 0.60 = 120 bpm*
- *70%: 200 × 0.70 = 140 bpm*
- *80%: 200 × 0.80 = 160 bpm*
- *90%: 200 × 0.90 = 180 bpm*

14. Write one healthy living practice related to sleep [An] / [CO2]

A healthy living practice related to sleep is maintaining a regular sleep pattern. Adequate and consistent sleep is essential for both physical and mental well-being. During sleep, the body repairs and regenerates

tissues, builds muscles, and strengthens the immune system. It also helps consolidate memories and improve concentration, learning, and decision-making. A proper sleep schedule helps regulate hormones responsible for appetite, growth, and stress control. Adults should aim for about 7 to 9 hours of sleep each night and avoid caffeine, heavy meals, or screen time before bedtime. Maintaining a regular sleep routine enhances mood, energy levels, and overall quality of life, making it a key element of healthy living.

15. Explain the role of physical inactivity in causing obesity [U] / [CO2]

Physical inactivity plays a major role in the development of obesity. When a person does not engage in sufficient physical activity, fewer calories are burned, leading to an energy imbalance where calorie intake exceeds expenditure. This excess energy is stored in the body as fat. Inactivity also slows down metabolism and promotes the accumulation of visceral fat, which surrounds vital organs and contributes to serious health risks. Moreover, a sedentary lifestyle leads to muscle loss, which further reduces energy expenditure and increases fat storage. Over time, this cycle contributes to obesity and increases the risk of associated lifestyle diseases such as diabetes, hypertension, and heart disease. Regular physical activity is therefore essential for maintaining a healthy weight and overall well-being.

16. Differentiate between normal bone density, osteopenia, osteoporosis, and severe osteoporosis. [U] / [CO3]

Bone density reflects the strength and mineral content of bones, which determine their resistance to fractures. Normal bone density refers to healthy bones with a T-score of -1.0 or above, indicating adequate bone strength. Osteopenia occurs when bone density begins to decrease slightly, with a T-score between -1.0 and -2.5. It represents an early warning stage where bones start to lose minerals and become weaker. Osteoporosis is a more advanced condition, defined by a T-score of -2.5 or below, where bone loss becomes significant and the risk of fractures increases. Severe osteoporosis occurs when the T-score is -2.5 or below along with one or more fractures, indicating extreme fragility. Maintaining bone health through regular weight-bearing exercises, sufficient calcium and vitamin D intake, and an active lifestyle can help prevent osteoporosis and maintain bone strength throughout life.

17. Discuss Squats with Loop and their importance for glute strength ? [E] / [CO5]

Squats with a loop (resistance band) are an effective lower-body exercise used to strengthen the gluteal muscles. In this exercise, a resistance loop is placed around the thighs just above the knees, and the person performs squats while maintaining outward tension on the band. This movement activates the gluteus maximus, medius, and minimus,

improving muscle tone, strength, and endurance. The added resistance from the loop increases the intensity of the workout and helps in better muscle engagement. Regularly performing squats with a loop enhances posture, balance, and lower-body stability. Strong glutes also help in supporting the spine and pelvis, reducing the risk of injuries in the lower back and knees. Thus, squats with loops are an excellent functional exercise for developing glute strength and overall lower-body fitness.

18. Write a short note on factors affecting RDA. [U] / [CO6]

Age, gender, activity level, physiological state (pregnancy, lactation), and environmental conditions.

[5x4 = 20]

Part D

ESSAY QUESTION

Answer any 1 questions

Each question carries 15 marks

19. Discuss modern lifestyle habits and their impact on chronic diseases. [E] / [CO6]

Common examples include diabetes, hypertension (high blood pressure), heart disease, obesity, cancer, and arthritis. These diseases often require ongoing medical care and lifestyle changes to manage symptoms and prevent complications.
Causes: Less exercise, junk food, stress, poor sleep, smoking, alcohol.
Effects: Obesity, diabetes, heart disease, hypertension.
Prevention: Exercise, eat healthy, sleep well, reduce stress, avoid bad habits.

20. What are psychosomatic disorders, and how do stress, anxiety, and strong emotions contribute to their development? [U] / [CO4]

Psychosomatic disorders are physical illnesses caused by mental or emotional stress. The word psychosomatic means a connection between the mind and body. In these disorders, emotional problems show up as physical symptoms.
These are not imaginary—the symptoms are real and can affect daily life. Common examples include headache, ulcers, back pain, asthma, and high blood pressure. Stress is one of the main causes of psychosomatic disorders. When a person is stressed, the brain releases stress hormones like adrenaline and cortisol. These hormones increase heart rate, blood pressure, and muscle tension. Long-term stress can lead to chronic diseases such as hypertension and heart problems. Anxiety also contributes by keeping the body in a constant “alert” state. People with anxiety may feel chest tightness, stomach upset, or dizziness. Strong emotions like anger, sadness, or fear can disturb normal body functions. Suppressed emotions may cause fatigue, digestive issues, or skin

problems. Managing stress through yoga, meditation, or exercise can reduce symptoms. A healthy lifestyle, proper sleep, balanced diet, and emotional support help prevent psychosomatic disorders.

21. Discuss in detail the physical, mental, and emotional benefits of regular physical activity ? [An] / [CO3]

Physical Benefits

- Improves cardiovascular health: Strengthens the heart, enhances blood circulation, and reduces the risk of heart disease and stroke.
- Enhances muscular strength and endurance: Promotes muscle tone, flexibility, and joint stability.
- Regulates body weight: Helps burn calories, prevents obesity, and maintains a healthy body composition.
- Boosts immunity: Strengthens the immune system and helps the body fight infections.
- Improves bone health: Increases bone density, preventing osteoporosis and age-related bone loss.

Mental Benefits

- Reduces stress and anxiety: Physical activity releases endorphins that help calm the mind and reduce tension.
- Enhances cognitive function: Improves memory, concentration, and overall brain performance.
- Improves sleep quality: Promotes deeper, more restful sleep by regulating biological rhythms.
- Increases self-confidence: Achieving fitness goals and feeling healthy improves self-esteem.
- Prevents mental health disorders: Regular exercise lowers the risk of depression, anxiety, and cognitive decline.

Emotional Benefits

- Promotes positive mood: Physical activity triggers the release of serotonin and dopamine, creating feelings of happiness.
- Improves emotional resilience: Helps individuals handle stress and emotional challenges effectively.
- Encourages social interaction: Group or team activities build connections and reduce loneliness.
- Enhances motivation and discipline: Regular exercise develops goal-setting habits and perseverance.
- Boosts overall life satisfaction: Contributes to a sense of well-being, balance, and inner peace.

[15x1 = 15]